MIL-F-46410A(Ord)
28 June 1962
SUPERSEDES
MIL-F-46410 (Ord)
18 October 1960

MILITARY SPECIFICATION
FOR
FUZE, MT, M-542, ASSEMBLY
AND
FUZE, MT, M-543A1, LESS BOOSTER ASSEMBLY

1. SCOPE

1.1 This specification covers one type of time functioning fuze for use with large caliber artillery ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bid form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	Ammunition and Special Weapons, General Specification for
MIL-I-45208	Inspection Requirements, General Specification for
MIL-P-14100	Primers, Stab Type, Loading, Assembling and Packing

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Inspection Terms and Definitions
MIL-STD-350	Jolt Test for Use in Production of Fuzes
MIL-STD-351	Jumble Test for Use in Production of Fuzes
MIL-STD-352	Forty (40) Foot Drop Test for Use in Production of Fuzes
MIL-STD-353	Transportation-Vibration Test for Use in Production of Fuzes
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-639	Surface Roughness Values Preferred for Army Ordnance Material

FSC 1390

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DRAWINGS

Ordnance Corps

F-8594464	Fuze, MT, M542, Assembly
F-8594380	Fuze, MT, M543A1, Less Booster Assembly
IEL-8594464	Inspection Equipment List for Fuze, MT, M542 Assembly
IEL-8594380	Inspection Equipment List for Fuze, MT, M543A1 Less Booster Assembly

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Dimensions and tolerances. - Dimensions and tolerances which affect external profile or which affect assembly with components external to the fuze are mandatory. Inspection requirements for these dimensions and tolerances are included under section 4 of this specification. Other dimensions and tolerances referred to in paragraph 6.4 will be controlled in accordance with MIL-I-45208; however, all other requirements of drawings and specification including basic design and geometry of each component part are mandatory.

3.1.1 Deviations. - Dimensions and tolerances which deviate from drawings shall be recorded on the basis of applicability to each fuze lot submitted for acceptance (see 6.4).

3.2 General. - Materials, parts, and assemblies shall comply with the requirements specified on the applicable drawings, standards, and specifications except as provided for in 6.4.

3.3 Initial production sample. - Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.4 Dynamic regulation. - The movement sub-assembly shall be regulated to the frequency specified on the applicable drawing when rotated at 6000 RPM plus or minus 25 RPM.

3.5 Jolt. - The fuze shall withstand the jolt when tested in accordance with Standard MIL-STD-350.

3.6 Jumble. - The fuze shall withstand the jumble when tested in accordance with Standard MIL-STD-351, except that the fuze shall be jumbled in a simulated metal packing container.

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3.7 Forty (40) foot drop. - The fuze shall withstand the drop when tested in accordance with Standard MIL-STD-352, except that the fuze shall be dropped in a simulated metal packing container.

3.8 Transportation - vibration. - The fuze shall withstand the vibration when tested in accordance with Standard MIL-STD-353.

3.9 Safety and spin fire

3.9.1 Six (6) foot drop and spin fire. - The fuze shall not fire when dropped six feet, base down in a protecting container. After the drop, the fuze shall fire when rotated at 2500 RPM plus or minus 15 RPM (see 4.4.12).

3.9.1.1 Explosive train. - The explosive train for each timer in the assembled fuze shall be independently capable of supplying sufficient energy to detonate the igniter or relay, as applicable.

3.10 Ballistics. - The fuze shall be disassembled to separate the timers. The timers shall be fired in a test shell and shall comply with the requirements of table II, as applicable (see 4.3.3.12).

3.10.1 Premature. - There shall be no premature bursts (see 6.1).

3.11 Workmanship. - The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.11.1 Metal defects. - All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.11.2 Firing arm assembly. - The firing arm shall contain no fracture of the bend.

3.11.3 Threads. - Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing and the provisions of Specification MIL-A-2550.

3.11.4 Burr. - No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.11.5 Foreign matter. - No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspection set forth in the specification where such inspection is deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection terms and definitions. - Reference shall be made to Standard MIL-STD-109 to define the inspection terms used.

4.1.2 Inspection. - Inspection shall be in accordance with Specifications MIL-A-2550 and MIL-I-45208 except as specified herein.

4.2 Initial production sample. - Prior to the initiation of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system and other inspection requirements of Specification MIL-I-45208. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.2.1 M542 Assembly. - Fifty-two fuze assemblies shall be tested as follows: Failure of any fuze to comply with the requirement as applicable, shall be cause for rejection of the initial production sample.

Test	Quantity	Requirement Paragraph
Jolt	4	3.5
Jumble	4	3.6
Forty (40) Foot Drop	5	3.7
Transportation-Vibration	5	3.8
Six Foot Drop	6	3.9.1, 3.9.1.1
Total	24	

4.2.1.1 Ballistic, M542. - The Government will perform ballistic tests on 28 fuzes, consisting of 84 timers. The acceptance criteria shall be in accordance with table II and paragraph 4.3.3.12.

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4.2.2 M543A1 Assembly. - Forty-eight fuze assemblies shall be tested as follows: Failure of any fuze to comply with the requirement as applicable, shall be cause for rejection of the initial production sample.

Test	Quantity	Requirement Paragraph
Jolt	4	3.5
Jumble	4	3.6
Forty (40) Foot Drop	5	3.7
Transportation-Vibration	5	3.8
Six Foot Drop	6	3.9.1, 3.9.1.1
Total	24	

4.2.2.1 Ballistic M543A1. - The Government will perform ballistic tests on 24 fuzes, consisting of 48 timers. The acceptance criteria shall be in accordance with table II and paragraph 4.3.3.12.

4.2.3 M542 or M543A1 assembly. - A sample of 25 each sub-assembly listed in paragraph 4.3.3.2.1 and 4.3.3.2.2 shall be submitted for static load and torque tests. Acceptance criteria shall be in accordance with paragraph 4.3.3.2.

4.3 Inspection provisions

4.3.1 Lot

4.3.1.1 Submittal of product. - The contractor shall submit the product in accordance with Standard MIL-STD-105 or MIL-STD-414 as applicable.

4.3.2 Examination. - One hundred percent examination shall perform for all critical defects. Examination for major and minor defects shall be performed on a defects basis in accordance with Standard MIL-STD-105. The criteria in table I herein shall be used for the classification of defects. All nonconforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed in Standard MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE I

Defect Classification	
Major	AQL - 0.25 percent
Minor	AQL - 0.40 percent

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4.3.2.1 Firing arm assembly. - Dwg. C-8593094 covering a detail of Dwgs. F-8594464 and F-8594380.

Defect	Method of Inspection
CRITICAL	
1. Fracture of upright bend of firing arm	Visual
MAJOR	
101. Part missing	Visual
MINOR	
None defined	

4.3.2.2 Delay arming mechanism assembly (Prior to winding of assembly) Dwg. F-8594387 covering a detail of Dwgs. F-8594380 and F-8594464.

CRITICAL	
None defined	
MAJOR	
101. Red end of detonator not at top of rotor (Plate pins up)	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.3.2.3 Delay arming mechanism assembly - Dwg. F-8594387 covering a detail of Dwgs. F-8594464 and F-8594380.

CRITICAL	
None defined	
MAJOR	
101. Insecure assembly	Manual
102. Rotor drive gear assemblies not in fully wound position (to be viewed through peephole in top plate)	Visual

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Defect	Method of Inspection
MAJOR (Cont'd)	
103. "D" shaft pin not in proper position (to be viewed through peephole in top)	Visual
104. Escapement lever not locked on proper side	Visual
105. Improper assembly	Visual
106. Part missing	Visual
107. Rotor retaining spring, damaged or deformed	Visual
108. "D" shaft spring, damaged or deformed	Visual
109. Staking missing or inadequate	Visual
110. Foreign matter	Visual
111. Burr	Visual
MINOR	
None defined	

4.3.2.4 Movement sub-assembly (after dynamic regulation) Dwg. F-8593032 covering a detail of Dwgs. F-8594464 and F-8594380.

CRITICAL	
1. Firing arm fires on "not fire" position	Gage
MAJOR	
101. Minimum protrusion of firing pin (in fired position)	Gage
102. Insecure assembly	Manual
103. Mainspring not engaged with anchor pin	Manual
104. Movement not fully wound (to be viewed through top plate for alignment of hole in lever assembly and hole in top plate)	Visual
105. Part missing	Visual
106. Improper assembly	Visual
107. Firing arm shaft spring protrudes beyond O.D. of movement	Visual
108. Metal defective	Visual
109. Foreign matter	Visual
110. Burr	Visual
MINOR	
None defined	

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4.3.2.5 Movement assembly, after assembly of timing disc assembly, (prior to assembly of locknut and retainer) Dwg. F-8594465 covering a detail of Dwgs. F-8594464 and F-8594380.

Defect	Method of Inspection
CRITICAL	
1. Timing disc bushing loose on knurl of arbor (to be checked prior to assembly of locknut)	Manual
MAJOR	
101. Metal defective	Visual
MINOR	
None defined	

4.3.2.6 Movement assembly - Dwg. F8594465 covering a detail of dwgs. F-8594464 and F-8594380.

CRITICAL	
1. Timing disc retainer, insecure	Manual
MAJOR	
101. Insecure assembly	Manual
102. Edge of firing arm not in alignment with lines on safety disc	Visual
103. Part missing	Visual
104. Improper assembly	Visual
105. Escapement lever not locked on proper side	Visual
106. Foreign matter	Visual
107. Burr	Visual
MINOR	
None defined	

4.3.2.7 Timers, numbers 1, 2 and 3 sub-assemblies Dwgs. D-8594470, D-8594471, and D-8594463, covering details of Dwgs. F-8594464 and F-8594380.

CRITICAL

None defined

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4.3.2.7 Cont'd

Defect	Method of Inspection
MAJOR	
101. Concentricity of gear and arbor assembly with datum A	Gage
102. Perpendicularity of gear and arbor assembly with datum B	Gage
103. Height of arbor, max.	Gage
104. Insecure assembly	Manual
105. Improper assembly	Visual
106. Metal defective	Visual
107. Foreign matter	Visual
108. Burr	Visual
109. Varnish missing or improperly applied	Visual
MINOR	
None defined	

4.3.2.8 Timers, numbers 1, 2 and 3 assemblies, Dwg. F-8594397 and F-8594461 covering details of Dwg. F-8594464 and F-8594380.

CRITICAL	
1. Safety disc off safe position (to be checked after tensioning lockwire)	Visual
MAJOR	
101. Setting pin insecure	Manual
102. Improper assembly	Visual
103. Timing disc zero setting incorrect	Visual
104. Timing disc disengaged from setting pin	Visual
105. Part missing	Visual
106. Body assembly screws missing	Visual
107. Zero line on body not aligned with zero line on housing	Visual
108. Metal defective	Visual
109. Foreign matter	Visual
110. Burr	Visual
111. Cement missing from specified locations	Visual
MINOR	
201. Staking missing or inadequate	Visual
202. Setting line not clearly visible	Visual

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4.3.2.9 Fuze Assembly - Dwg. F-8594464 for the MT, M-542 and Dwg. F-8594380 for the MT, M-543A1 less Booster Assembly.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of large outside thread	Gage
102.	Pitch diameter of small outside thread (M-543A1)	Gage
103.	Major diameter of large outside thread, min	Gage
104.	Major diameter of small outside thread, min (M-543A1)	Gage
105.	Pitch diameter of internal thread (M-542)	Gage
106.	Minor diameter of internal thread (M-542)	Gage
107.	Concentricity of housing with large outside thread	Gage
108.	Diameter top, max.	Gage
109.	Total length, max.	Gage
110.	Insecure assembly	Manual
111.	Pellet holder assembly missing (to be checked prior to bolting housing to breech block)	Visual
112.	Charge holder assembly missing (M-543A1)	Visual
113.	Gasket(s) missing in flash tube (to be checked during assembly)	Visual
114.	Zero line on housing not aligned with safe line on dial	Visual
115.	Marking missing, incorrect or illegible	Visual
116.	Part missing, except as otherwise classified	Visual
117.	Thread damaged or not full	Visual
118.	Metal defective	Visual
119.	Foreign matter	Visual
120.	Burr	Visual
MINOR		
	None defined	

4.3.2.10 Packing and Marking - Inspection for packing and marking shall be as directed by the contracting officer.

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4.3.3 Inspection testing - The following tests shall be performed in accordance with the provisions of 4.4 except as provided in Specification MIL-I-45208. Referenced drawings cover details of Drawings F-8594380 and F-8594464.

4.3.3.1 Protective finish - The number of units for protective finish test shall be in accordance with the applicable specification on the component drawings. Failure to comply with the requirements shall be cause for rejection of the lot of components.

4.3.3.2 Static load and torque - The sample size and criteria for acceptance shall be as specified for range plans in Standard MIL-STD-414, Form 2. An AQL of 0.25 percent shall be used. If the variables analysis indicate rejection but no individual value has actually failed to meet the drawing requirements, a comparable attribute plan in Standard MIL-STD-105 shall be used, in which case the sample units used for the variables tests shall be considered a part of the attributes sample.

4.3.3.2.1 Static Load -

	Drawing
Pallet Lock Assembly	B-8593016
Pallet Lock Assembly	B-8593152
Disc and Stud Assembly	C-8593103
Escapement Gear and Pinion Assembly (Movement)	C-8593382
Escapement Gear and Pinion Assembly (Delay Mech)	C-8594411
Escapement Lever Assembly (Movement)	F-8593015
Escapement Lever Assembly (Delay Mech)	D-8594406
Firing Arm Assembly (Test of Pin)	C-8593094
Firing Arm Assembly (Test of Arm)	C-8593094
Lever Assembly	B-8593124
No. 1 Gear and Pinion Assembly (Movement)	C-8594280
No. 1 Gear and Pinion Assembly (Delay Mech)	C-8594418
No. 2 Gear and Pinion Assembly (Movement)	C-8594283
No. 2 Gear and Pinion Assembly (Delay Mech)	C-8594419
No. 3 Gear and Pinion Assembly (Movement)	C-8594286
No. 3 Gear and Pinion Assembly (Delay Mech)	C-8594420
No. 4 Gear and Pinion Assembly	C-8593085
No. 5 Gear and Arbor Assembly	C-8593090
Rotor Assembly	D-8594410
"D" Shaft Assembly	C-8594415
Centrifugal Drive Gear Assembly	C-8594473

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4.3.3.2.2 Torque -

	Drawing
Disc and Stud Assembly	C-8593103
Escapement Lever Assembly (Movement)	F-8593015
Escapement Lever Assembly (Delay Mech)	D-8594406
No. 5 Gear and Arbor Assembly	C-8593090

4.3.3.3 Hardness - Twenty samples from each lot of the following components shall be subjected to the hardness test. Failure of any component to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot.

Component	Drawing
Arm, Firing	C-8593095
Spring, Timing Disc	B-8594454
Guide, Adjusting	B-8594273
Lever, Escapement (Movement)	F-8593023
Lever, Escapement (Delay Mech)	D=8594400
Mainspring	B-8593112
Pin, Firing Arm Shaft	B-8593022

4.3.3.4 Bend, Firing arm - The same 20 components used in 4.3.3.3 from each lot that comply with the hardness requirements shall be subjected to the bend test. Failure of any component to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot.

4.3.3.5 Dead load, Timing disc spring - The same 20 components used in 4.3.3.3 from each lot that comply with the hardness requirements shall be subjected to the dead load test. Failure of any component to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot.

4.3.3.6 Compression, Firing pin spring - Twenty springs from each lot shall be subjected to the compression test. Failure of any spring to comply with the requirements of drawing B8594315 shall be cause for rejection of the lot.

4.3.3.7 Kinetic energy, Firing pin - Twenty movement sub-assemblies from each lot shall be subjected to the kinetic energy test. Failure of any sub-assembly to comply with the requirements of the applicable drawing shall be cause for rejection of the lot.

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4.3.3.8 Arming and Non-arming (Movement sub-assembly) - Each movement sub-assembly shall be subjected to the arming and non-arming tests. Failure of any sub-assembly to comply with the requirements of the applicable drawing shall be cause for rejection of the unit.

4.3.3.9 Arming and Non-arming (Delay arming mechanism assembly) - Each delay arming mechanism assembly shall be subjected to the arming and non-arming tests. Failure of any assembly to comply with the requirements of the applicable drawing shall be cause for rejection of the unit.

4.3.3.10 Hammer spring deflection (Upper, middle and lower dial assemblies) - Each dial assembly shall be subjected to the hammer-spring deflection test. Failure of any assembly to comply with the requirements of the applicable drawing shall be cause for rejection of the unit.

4.3.3.11 M542 or M543A1 Assembly - Eleven fuze assemblies from each lot, as applicable shall be tested as follows: Failure of any assembly to comply with the requirement as applicable shall be cause for rejection of the lot.

Test	Quantity	Requirement Paragraph
Transportation-Vibration	5	3.8
Six Foot Drop	6	3.9.1, 3.9.1.1

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4.3.3.12 Ballistic - The Government will perform ballistic tests on 28 fuzes, consisting of 84 timers for the M542 fuze, or 24 fuzes, consisting of 48 timers for the M543A1 fuze. The occurrence of any of the following shall be cause for rejection of the lot which the ballistic sample represents.

a. A premature burst (see 6.1)

b. An off-time, fast (see 6.3)

c. More than 2 off-times, slow (see 6.3)

d. More than 2 time failures (see 6.2)

e. Failure to comply with the requirement of 3.10, except as permitted in c. and d. above.

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TABLE II

Fuze	Weapon	Zone	Timer Setting Seconds	Limits for Ind. Time to Burst Seconds		Limits for Avg. Time to Burst Seconds		Std. Dev. Max.	"Timer" Sample Size
				LLX	ULX	$LL\bar{X}$	$UL\bar{X}$		
M-542	8" Howitzer	I	20	19.72	20.95	20.15	20.52	.23	38, 39, 40, 41, 42
M-542	8" Howitzer	VII	55	54.24	56.09	54.94	55.39	.37	38, 39, 40, 41, or 42
M-543A1	8" Howitzer	I	20	19.72	20.95	20.12	20.54	.24	20, 21, 22, 23 or 24
M-543A1	8" Howitzer	VII	55	54.24	56.09	54.91	55.42	.39	20, 21, 22, 23 or 24

Note:

The time-to-burst shall be recorded.

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4.3.3.13 Process control - Periodic unscheduled monitoring shall be conducted for the following process operations to determine compliance with the applicable requirements of the indicated drawings.

Lubrication	F8593032
"	F8593043
"	F8594387
"	D8594404
Lock nut (torque)	F8594465
Timing Disc (torque)	F8594465
Timing Disc Assembly (torque)	F8594472
Staking top and bottom screws to assembly, prior to dynamic regulation of movement	F8593032
Torque between dial and housing	
Timer No. 1 Assembly (if applicable)	F8594396
Timer No. 2 Assembly	F8594397
Timer No. 3 Assembly	F8594461

4.3.3.14 Surveillance of Primers - Each lot of T103E1 primers shall be surveillance tested for sensitivity in accordance with methods and acceptance criteria specified in MIL-P-14100 and Drawing D-8787058 covering a detail of Drawings F-8594464 and F-8594380. Lots subjected to this test within six (6) months prior to final assembly of fuze need not be tested. Primer lots which fail to pass the acceptance criteria shall not be used.

4.3.4 Inspection Equipment - The examination and tests shall be made using equipment listed on IEL-8594464 for fuze MT, M-542, and IEL-8594380 for fuze, MT, M-543A1, except as provided in Specification MIL-I-45208.

4.4 Test Methods and Procedures

4.4.1 Static Load - Samples shall be tested to the point of disassembly. The values obtained shall be used to determine lot acceptability in accordance with the specified variable plan. Escapement lever assemblies shall be tested prior to insertion of the escapement lever spring. All samples tested shall be scrapped.

4.4.2 Torque - Samples shall be tested to the breakaway point. The values obtained shall be used to determine lot acceptability in accordance with the specified variable plan. Escapement lever assemblies shall be tested prior to insertion of the escapement lever spring. Escapement lever assemblies and No. 5 gear and arbor assemblies tested shall be scrapped.

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4.4.3 Hardness - Samples shall be tested in an area which is solidly supported in order to obtain valid readings. The average value of 3 hardness readings taken on each sample unit shall be used as the individual value.

4.4.4 Firing arm bend - Test procedures shall be in accordance with the applicable drawing. Firing arms subjected to the bend test shall be scrapped.

4.4.5 Timing disc spring dead load - Test procedures shall be in accordance with the applicable drawing. Timing disc springs subjected to the dead load test shall be scrapped.

4.4.6 Firing pin spring compression - Test procedures shall be in accordance with the applicable drawing. Firing pin springs subjected to the compression test shall be scrapped.

4.4.7 Kinetic energy - The movement sub-assembly shall be tested for the firing pin kinetic energy by releasing the firing pin from the cocked position.

4.4.8 Jolt - The fuze shall be jolted in accordance with MIL-STD-350, except that the fuze shall not be jolted in the horizontal position. After disassembly and examination, all components shall be scrapped. (Applies to initial production sample only).

4.4.9 Jumble - The fuze shall be jumbled in accordance with Standard MIL-STD-351. After disassembly and examination, all components shall be scrapped. (Applies to initial production sample only).

4.4.10 Forty (40) foot drop - The fuze shall be dropped in accordance with Standard MIL-STD-352. After disassembly and examination, all components shall be scrapped. (Applies to initial production sample only).

4.4.11 Transportation-Vibration - The fuze shall be tested in accordance with MIL-STD-353. The vibrator used shall be the Naval Ordnance Lab, Type 2, or any commercial vibrator which permits compliance with the test procedures of MIL-STD-353. After disassembly and examination, all components shall be scrapped.

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4.4.12 Six (6) foot arming drop and spin fire - The fuze shall be set at 15 second intervals between 12 and 87 seconds. An adapter approved by the contracting officer, shall replace the breech block of each fuze to determine whether the individual timers are capable of supplying sufficient energy to detonate the igniter. Each fuze shall be assembled in a protecting container and dropped base down from a height of 6 feet in such a manner as to fall freely and impact upon a lead plate 3/4 inch $\pm$ 1/32 inch thick. The plate shall be made of lead antimony alloy containing 10 percent maximum by weight of antimony, cast in a confining ring of steel to prevent the free flow of the alloy under impact. The lead antimony plate shall be placed on a solidly supported steel plate at least 2 inches thick. Immediately upon completion of the 6 foot arming drop, the same fuzes shall be removed from their containers and assembled to a spin chuck and rotated in accordance with 3.9. The set time and the time to fire shall be recorded for each fuze. Should any timer in the fuze fail to fire after spinning, it shall be disassembled and examined in an effort to ascertain the cause. After disassembly and examination, all components shall be scrapped.

4.4.13 Time to Arm (Delay arming mechanism assembly) - The delay arming mechanism assembly shall be placed in a spin fixture and spun at speeds specified on the applicable drawing. The time required to arm the assembly shall be recorded for each spin. (Prior to and at the conclusion of each spin the assembly shall be fully wound.) The time to arm test shall be performed prior to the non-arming test.

4.4.14 Non-arming of delay arming mechanism assembly - The non-arming test shall be performed on completely wound delay arming mechanism assemblies. The delay mechanism assemblies shall be rotated at the RPM specified on the applicable drawing for a time interval sufficient to determine conformance with the requirement.

4.4.15 Arming of movement sub-assembly - The arming test shall be performed on completely wound movement sub-assemblies. The sub-assemblies shall be rotated at the RPM specified on the applicable drawing for a time interval sufficient to determine conformance with the requirement. At the conclusion of the test, each mechanism shall be rewound. The arming test shall be performed prior to the non-arming test.

4.4.16 Non-arming of movement sub-assembly - The non-arming test shall be performed on completely wound sub-assemblies. The sub-assemblies shall be rotated at the RPM specified on the applicable drawing for a time interval sufficient to determine conformance with the requirement.

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4.4.17 Hammerspring deflection - (Upper Dial, Middle Dial, and Lower Dial Assemblies) - The required load shall be applied at the specified location parallel to the vertical axis of the assembly.

4.4.18 Dynamic regulation - Each movement shall be dynamically regulated with an approved regulating machine. The spinning chuck shall rotate at a constant speed of 6000 ± 25 RPM. Timekeeping and performance of the movement shall be checked over a period of not less than 75 seconds. No adjustment shall be made to the escapement during the 75 second check run. Should any movement require more than 4 check runs, it shall be rejected, subject to the insertion of a new escapement spring. The replaced escapement spring shall be scrapped.

4.4.19 Ballistic tests - The fuzes shall be disassembled by removing the 3 assembly bolts and the breech block from each fuze assembly. The identity of the timers shall be maintained for each fuze. For the MT, M-542 Fuze, one group of 42 timers shall be fired in the 8" Howitzer, Zone I, setting 20 seconds. The remaining 42 timers shall be fired in the 8" Howitzer, Zone VII, setting 55 seconds. For the MT, M-543A1 fuze, one group of 24 timers shall be fired in the 8" Howitzer, Zone I setting 20 seconds. The remaining 24 timers shall be fired in the 8" Howitzer, Zone VII, setting 55 seconds. Each timer for each model fuze shall be assembled to the test shell as a complete round. The boosters shall be selected from an accepted lot of boosters. All settings shall be made using fuze setter utilizing the vernier scale on the fuze housing for accurate setting. Fuze chronograph and electric clock time to burst shall be recorded. The average and the standard deviation shall be calculated for each group of timers. Any timer which is an off time, or any timer which fails to function shall be excluded from the computation. (See 6.2 and 6.3).

5. PREPARATION FOR DELIVERY

5.1 General - When the booster or igniter assembly is assembled to the fuze at the contractor's plant, the packaging requirements contained in the complete item specification are applicable.

5.2 Packing and Marking -

5.2.1 Level C -

5.2.1.1 Packing and marking shall be in accordance with the terms of the contract or as directed by the contracting officer.

5.3 Data cards - Data card information shall be as outlined in Specification MIL-A-2550.

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6. NOTES

6.1 A premature burst is considered to have occurred if any of the following events occur:

a. If the round detonates in the bore of the gun.

b. If the time of flight is less than 75 percent of the set time.

c. If the time of flight is less than 8.6 seconds.

6.2 A time failure is considered to have occurred if the round fails to detonate during flight.

6.3 An off time, either fast or slow, is any individual air burst occurring outside the limits established for individual times of flight.

6.4 Dimensions and tolerance represent a design objective for the manufacture of fuzes. It is recognized that many combinations of dimensions and tolerances will result in an acceptable fuze. Therefore, compliance with specified dimensions and tolerances other than those specified in paragraph 3.1 is not mandatory.

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